

# Nirma University

## Institute of Technology

Semester End / Supplementary Examination (IR/SPE), February - 2022

Master of Computer Application, Semester - I

3MCA101 Object Oriented Programming

Roll /  
Exam No.

Supervisor's initial  
with date

Time: 2 Hours

Max. Marks: 50

Instructions:

1. Attempt all questions.
2. Figures to right indicate full marks.
3. Draw neat sketches wherever necessary.
4. Assume suitable data wherever necessary and mention the same.

**Q-1**      **Answers the following.**      **[18]**

- (A)      What is the role of the Java Development Kit (JDK) in Java (6)  
**CO1,BL2**      programming language? Explain it in detail.
- (B)      Explain various Relational operators. Also give an examples of (6)  
**CO1,BL5**      relational operators using java programming.
- (C)      Briefly describe JDBC (Java Database Connectivity). Discuss the (6)  
**CO3,BL6**      contribution of the JDBC in development of client server application?

**OR**

- (C)      What is a stream in context of Java I/O? What types of streams (6)  
**CO3,BL6**      does Java define? Give two example of Java built in streams.

**Q-2**      **Answers the following.**      **[14]**

- (A)      Briefly describe Constructor. Differentiate between the Java (7)  
**CO2,BL4**      Constructor and Java method by giving a suitable example.
- (B)      Write an importance of the loop in Java programming? Also give (7)  
**CO2,BL5**      an example of each types of the loop used in Java programming.

**Q-3**      **Answers the following.**      **[18]**

- (A)      Is it compulsory to use exception handling in Java programming? (9)  
**CO3,BL5**      Explain the try and catch block with a suitable example which will use an ArithmeticException.
- (B)      Discuss various types of inheritance. Write a java program which (9)  
**CO2,BL6**      will show the use of the multilevel inheritance.

**OR**

- (B)      Discuss the function of abstract class and interface with a (9)  
**CO2,BL6**      suitable example and also differentiate between them.